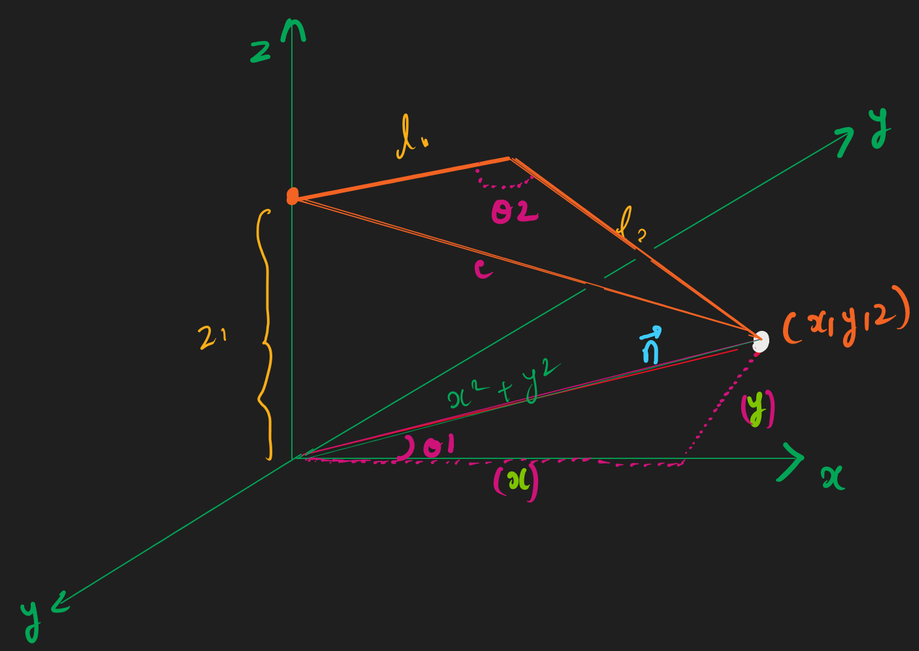
INVERSE KINEMATICS (2R) Robot

(.....) (⋮) perpendicular axis of Rotation



$$\boxed{c^2 = z_1^2 + x^2 + y^2} \quad (3)$$

$$\tan\theta_1 = \frac{y}{x}$$

$$\boxed{\theta_1 = \tan^{-1}(y/x)} \quad (1)$$

$$\boxed{\cos(\theta_2) = \frac{l_1^2 + l_2^2 - c^2}{2 \times l_1 \times l_2}}$$

Cosine Rule

$$\boxed{\theta_2 = \cos^{-1}\left[\frac{l_1^2 + l_2^2 - c^2}{2 \times l_1 \times l_2}\right]} \quad (2)$$